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1. (12 points) Answer each of the following questions precisely but shortly (no need of a proof).
- (a) (2 points) State the First Isomorphism Theorem.

**Solution:** Let  $\phi : G \rightarrow H$  be a group homomorphism. Then  $\ker(\phi)$  is a normal subgroup of  $G$ , and the map  $\bar{\phi} : G/\ker(\phi) \rightarrow \text{im}(\phi)$  defined by  $\bar{\phi}(g\ker(\phi)) = \phi(g)$  is a well-defined group isomorphism. That is,  $G/\ker(\phi) \cong \text{im}(\phi)$ .

- (b) (2 points) Recall that all subgroups  $H \leq \mathbb{Z}$  are all of the form  $H = n\mathbb{Z}$  for some nonnegative integer  $n$ . Then the subgroup  $n\mathbb{Z} + m\mathbb{Z} = d\mathbb{Z}$  for some integer  $d$ . What is  $d$ ?

**Solution:** The subgroup  $n\mathbb{Z} + m\mathbb{Z}$  is the set of all integer linear combinations of  $n$  and  $m$ , i.e.,  $\{an + bm : a, b \in \mathbb{Z}\}$ . This is precisely the subgroup generated by  $\gcd(n, m)$ , so  $d = \gcd(n, m)$ .

- (c) (2 points) What is the group  $\text{Aut}(\mathbb{Z}_p)$  when  $p$  is prime?

**Solution:** It is isomorphic to  $(\mathbb{Z}_p)^\times$ , the group of units in  $\mathbb{Z}_p$ , which is a cyclic group of order  $p - 1$ . In particular,  $\text{Aut}(\mathbb{Z}_p) \cong \mathbb{Z}_{p-1}$ .

- (d) (2 points) Give the definition of group action.

**Solution:** A *group action* of a group  $G$  on a set  $X$  is a map  $G \times X \rightarrow X$ , denoted  $(g, x) \mapsto g \cdot x$ , satisfying:

- (i) (Identity) For all  $x \in X$ ,  $e \cdot x = x$  where  $e$  is the identity element of  $G$ .
- (ii) (Compatibility) For all  $g, h \in G$  and  $x \in X$ ,  $(gh) \cdot x = g \cdot (h \cdot x)$ .

- (e) (2 points) State Burnside's lemma.

**Solution:** Let  $G$  be a finite group acting on a finite set  $X$ . For each  $g \in G$ , let  $\text{Fix}(g) = \{x \in X : g \cdot x = x\}$  be the set of elements of  $X$  fixed by  $g$ . Then the number  $k$  of orbits of  $G$  on  $X$  is given by

$$k = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|.$$

- (f) (2 points) How many **distinct** irreducible permutation representations of  $\mathfrak{S}_3$  are there? Take into account that the subgroups of  $\mathfrak{S}_3$  are  $H_1 = \{e\}$ ,  $H_2 = \langle(12)\rangle$ ,  $H_3 = \langle(13)\rangle$ ,  $H_4 = \langle(23)\rangle$ ,  $H_5 = \langle(123)\rangle$ , and  $H_6 = \mathfrak{S}_3$ .

**Solution:** By the correspondence between subgroups and transitive permutation representations, the irreducible permutation representations of  $\mathfrak{S}_3$  correspond to the conjugacy classes of subgroups. The distinct irreducible permutation representations are obtained from the subgroups:

- $H_1 = \{e\}$  gives the regular representation of cardinality  $6 = |\mathfrak{S}_3|$ .
- $H_5 = \langle(123)\rangle$  of order 3 gives a transitive representation of cardinality  $[S_3 : H_5] = 2$ .
- $H_2 = \langle(12)\rangle$  of order 2 gives a transitive representation of cardinality  $[S_3 : H_2] = 3$ .
- $H_6 = \mathfrak{S}_3$  gives the trivial representation of cardinality 1.

The conjugacy classes of subgroups of order 2 are all equivalent (since all involutions are conjugate in  $\mathfrak{S}_3$ ). Therefore, there are exactly **4** distinct irreducible permutation representations of  $\mathfrak{S}_3$ .

2. (5 points) Suppose that  $\mathcal{C}$  is a family of subsets of a group  $G$  which forms a partition of  $G$ , and suppose further that  $gC \in \mathcal{C}$  for any  $g \in G$  and  $C \in \mathcal{C}$ . Show that there is an  $H \in \mathcal{C}$  which is a subgroup of  $G$ . (Note that this is only part of the proof to show that  $\mathcal{C}$  is a set of left cosets.)

**Solution:** Let  $H$  denote the unique element of  $\mathcal{C}$  that contains the identity  $e$  (such an element exists because  $\mathcal{C}$  partitions  $G$ ). We will show that  $H$  is a subgroup of  $G$  and that  $\mathcal{C} = \{gH : g \in G\}$ .

*Closure:* Let  $h_1, h_2 \in H$ . Consider the set  $h_1H \in \mathcal{C}$ . Since  $h_1 = h_1 \cdot e \in h_1H$  and  $h_1 \in H$ , the two members  $h_1H$  and  $H$  of the partition  $\mathcal{C}$  share the element  $h_1$ . Because  $\mathcal{C}$  is a partition (its members are pairwise disjoint), we must have  $h_1H = H$ . In particular,  $h_1h_2 \in h_1H = H$ .

*Identity:*  $e \in H$  by definition.

*Inverses:* Let  $h \in H$ . From the closure argument we know  $hH = H$ . Since  $e \in H$ , there exists  $h' \in H$  such that  $hh' = e$ , i.e.,  $h^{-1} = h' \in H$ .

Therefore  $H \leq G$ .

3. (5 points) Assuming that we already know that  $\text{Inn}(G) \leq \text{Aut}(G)$  for any group  $G$ , show that  $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$ .

**Solution:** Let  $\varphi_g : G \rightarrow G$  be the inner automorphism defined by  $\varphi_g(x) = gxg^{-1}$  for any  $g \in G$ . Then  $\text{Inn}(G) = \{\varphi_g : g \in G\}$ .

To show that  $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$ , we must show that for any  $\varphi_g \in \text{Inn}(G)$  and any  $\sigma \in \text{Aut}(G)$ , the conjugate  $\sigma \circ \varphi_g \circ \sigma^{-1}$  is also in  $\text{Inn}(G)$ .

Let  $\sigma \in \text{Aut}(G)$  and  $g \in G$ . Then for every  $x \in G$ ,

$$(\sigma \circ \varphi_g \circ \sigma^{-1})(x) = \sigma(\varphi_g(\sigma^{-1}(x))) = \sigma(g\sigma^{-1}(x)g^{-1}) = \sigma(g) \cdot x \cdot (\sigma(g))^{-1} = \varphi_{\sigma(g)}(x).$$

So  $\sigma \circ \varphi_g \circ \sigma^{-1} = \varphi_{\sigma(g)}$ , which is an inner automorphism since  $\sigma(g) \in G$ . Therefore,  $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$ .

4. (5 points) Let  $G$  be a finite group and let  $H$  be a subgroup of  $G$  with  $[G : H] = p$ , where  $p$  is the smallest prime dividing  $|G|$ . Prove that  $H$  is normal in  $G$ .

**Solution:** Let  $G$  act on the set  $G/H$  of left cosets by left multiplication, i.e.,  $g \cdot (aH) = (ga)H$ . This gives a homomorphism  $\phi : G \rightarrow \mathfrak{S}_{G/H} \cong \mathfrak{S}_p$ , where  $\mathfrak{S}_p$  is the symmetric group on  $p$  elements.

Let  $K = \ker(\phi)$ . Since  $K$  is the kernel of a homomorphism,  $K \trianglelefteq G$ . We also know that  $K \subseteq H$ : indeed, if  $g \in K$  then  $g$  fixes every coset, and in particular  $g \cdot (eH) = gH = H$ , which means  $g \in H$ . So  $K \leq H \leq G$ .

By the First Isomorphism Theorem,  $G/K \cong \text{im}(\phi) \leq \mathfrak{S}_p$ , so  $|G/K|$  divides  $|\mathfrak{S}_p| = p!$ . That is,  $[G : K]$  divides  $p!$ .

On the other hand, by the tower law for indices,

$$[G : K] = [G : H] \cdot [H : K] = p \cdot [H : K].$$

Now  $[G : K]$  divides  $|G|$  (since  $K \leq G$ ) and  $[G : K]$  divides  $p!$ . Since  $p$  is the *smallest* prime dividing  $|G|$ , this can only happen if  $[G : K]$  divides  $p$ , and so  $[G : K]$  is either 1 or  $p$ . If  $[G : K] = 1$ , then  $K = G$ , which contradicts the fact that  $K \subseteq H$  and  $[G : H] = p > 1$ . Therefore, we must have  $[G : K] = p$  and  $[H : K] = 1$ , i.e.,  $K = H$ .

Since  $K = \ker(\phi) \trianglelefteq G$ , we conclude  $H \trianglelefteq G$ .

5. (5 points) **(Cayley's Theorem)** Prove that every group  $G$  is isomorphic to a subgroup of the symmetric group  $\mathfrak{S}_G$  (the group of all bijections  $G \rightarrow G$ ). In particular, if  $G$  is finite with  $|G| = n$ , then  $G$  is isomorphic to a subgroup of  $\mathfrak{S}_n$ .

**Solution:** Consider the left regular action of  $G$  on itself where  $G$  acts by left multiplication. This defines a group homomorphism  $\phi : G \rightarrow \mathfrak{S}_G$  where for every  $g \in G$  and every  $x \in G$ ,  $\phi(g)(x) = gx$ . The kernel of  $\phi$  is given by

$$\begin{aligned}\ker(\phi) &= \{g \in G : \phi(g)(x) = x \text{ for all } x \in G\} \\ &= \{g \in G : gx = x \text{ for all } x \in G\} \\ &= \{g \in G : g = e\} \\ &= \{e\}.\end{aligned}$$

We conclude that  $\phi$  is injective, and by the First Isomorphism Theorem,

$$G \cong G/\{e\} = G/\ker(\phi) \cong \text{im}(\phi) \leq \mathfrak{S}_G.$$

If  $|G| = n$ , we can identify  $\mathfrak{S}_G$  with  $\mathfrak{S}_n$ , and thus  $G$  is isomorphic to a subgroup of  $\mathfrak{S}_n$ .

6. (5 points) Let  $H$  and  $K$  be subgroups of a group  $G$ . Prove that if the product set  $HK$  is a subgroup of  $G$  then  $HK = KH$ . (Note that this is only one direction)

**Solution:** Recall  $HK = \{hk : h \in H, k \in K\}$  and  $KH = \{kh : k \in K, h \in H\}$ .

Assume  $HK$  is a subgroup of  $G$ . We show  $HK = KH$ .

( $\subseteq$ ) Let  $hk \in HK$ . Since  $HK \leq G$ , we have  $(hk)^{-1} = k^{-1}h^{-1} \in HK$ , so  $k^{-1}h^{-1} = h'k'$  for some  $h' \in H, k' \in K$ . Taking inverses of both sides:  $hk = (k')^{-1}(h')^{-1} \in KH$ . Hence  $HK \subseteq KH$ .

( $\supseteq$ ) Let  $kh \in KH$ . Then  $(kh)^{-1} = h^{-1}k^{-1} \in HK$ . Since  $HK$  is a subgroup,  $((kh)^{-1})^{-1} = kh \in HK$ . Hence  $KH \subseteq HK$ .

Therefore  $HK = KH$ .